METRIC

DOD-S-63275A
29 October 1982
SUPERSEDING
DOD-S-63275
26 September 1977

MILITARY SPECIFICATION

STUD, SCREW THREAD, LOCKED IN, RING LOCKED, SERRATED, METRIC AND RING, LOCK, SERRATED, METRIC, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metric locked in studs. The method of locking in the stud is by means of a serrated collar and an accessory lockring with matching serrations installed within the parent material to prevent rotation.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225/6 - Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn or Cold Finished, 2024
QQ-A-250/6 - Aluminum Alloy 5083, Plate and Sheet
QQ-P-416 - Plating, Cadmium (Electrodeposited)
PPP-H-1581 - Hardware (Fasteners and Related Items), Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5307

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MILITARY

MIL-I-6866 - Inspection, Penetrant Method of
MIL-I-6868 - Inspection Process, Magnetic Particle
MIL-H-6875 - Heat Treatment of Steels (Aircraft Practice, Process for)
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys
DOD-S-63275/1 - Stud, Screw Thread, Locked In, Ring Locked, Serrated, Metric

STANDARDS

FEDERAL

FED-STD-H28/20 - Inspection Methods for Acceptability of UN, UNR, UNJ, M and MJ Screw - Threads
FED-STD-H28/21 - Screw Thread Standards for Federal Services, Section 21, Metric Screw Threads

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1312 - Fasteners Test Methods

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B 46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AEROSPACE MATERIAL SPECIFICATIONS (AMS)

AMS 5662 Alloy Bars, Forgings and Rings, Corrosion and Heat Resistant (718)
AMS 6322 Steel, Alloy, Composition 8740

(Application for copies should be addressed th the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 380 - Cleaning and Descaling Stainless Steel Parts, Equipment and Systems
ASTM E 8 - Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references other than specification sheets cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials (see 3.1) shall be used to the maximum extent practicable. The studs shall be a one piece all metal unit, made from the following materials as specified.

3.2.1 Alloy steel. Alloy steel shall be composition 8740 (UNS G87400) in accordance with AMS 6322.

3.2.2 Nickel base alloy. Nickel base alloy shall be corrosion and heat resistant composition 718 (UNS N07718) in accordance with AMS 5662.

3.3 Treatment and finish.

3.3.1 Heat treatment. The stud shall be heat treated in accordance with the heat treatment process specified in MIL-H-6875.

3.3.2 Protective finish. The stud shall be furnished with a protective plating or surface treatment as specified herein.

3.3.2.1 Cadmium plate. Alloy steel studs shall be cadmium plated in accordance with QQ-P-416, type II, class 3. The plating thickness shall be 5.1 µm (micrometers).

3.3.2.2 Cleaning and descaling. Corrosion resistant steel studs shall be cleaned and descaled in accordance with ASTM A380.

3.4 Design, dimensions and tolerances. Design, dimensions, and tolerances shall conform to the requirements of the applicable specification sheet and shall apply after application of the protective plating or cleaning and descaling surface treatment.

3.4.1 Threads. Threads shall be right hand, M or MJ profile in accordance with FED STD H28/21 and the applicable specification sheet.

3.4.1.1 Nut end threads. Nut end threads shall be M or MJ profile, tolerance class 4h for alloy steel and 4e for nickel base alloy as specified in the applicable specification sheet.

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3.4.1.2 Stud end threads. Stud end threads shall be M profile, tolerance class 3h4h as specified in the applicable specification sheet.

3.4.1.3 Thread forming. Composition 8740 alloy steel stud threads shall be fully formed after heat treatment. Composition 718 nickle base alloy stud threads shall be formed prior to aging. Thread forming shall be prior to protective plating or surface treatment. Threads for nut end and stud end shall be formed by a single rolling process.

3.4.1.4 Incomplete threads. The runout threads shall be faired into the shank within a minimum of one and a maximum of two pitches without an abrupt change in cross sectional area. Lead thread shall not exceed two pitches. The root and flanks of the runout thread may deviate from true thread form but shall be smooth and free of tool marks.

3.4.1.5 Grain flow. The grain flow in rolled threads shall follow the general thread contour with the maximum density at the bottom of the root radius as shown in figure 1.

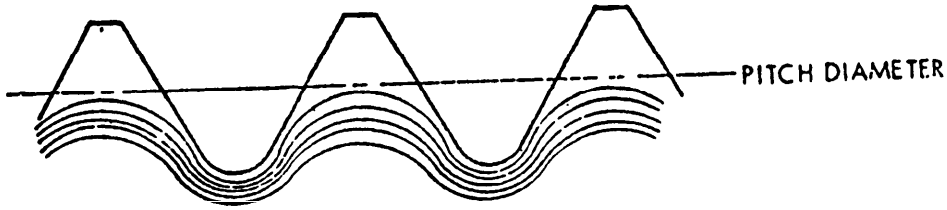


Figure 1. Thread Grain Flow

3.4.2 Locking serrations. Locking serration dimensions shall conform in size to the applicable specification sheet part number and shall be of full form and free from burrs or other flaws.

3.4.3 Surface texture. The surface texture, prior to plating, shall not exceed the values stated in the applicable specification sheet and shall be in accordance with ANSI B46.1.

3.4.4 Straightness. The straightness of the stud shall be within the values specified in table I when tested in accordance with 4.6.5.

TABLE I. Straightness

NUT END THREAD SIZE	DEVIATION OF STUD FROM PLATE, MAX. (mm PER 10mm OF STUD NUT END LENGTH)
M5	0.040
M6 THRU M8	0.030
M10	0.025
M12, M14	0.020

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3.5 Mechanical properties. Studs conforming to the design and dimensions specified in the applicable specification sheet and having load ratings as specified in Table II shall be capable of developing a minimum tensile strength, a minimum proof strength, and have shear engagement areas in accordance with table II.

TABLE II. Mechanical properties

Load Rating MPa	Min Tensile Strength MPa	Min Proof Strength MPa	Shear Engagement Area (Stud End)	
			-100 Numbers	-200 Numbers
900	900	630	TABLE V COL 1	TABLE V COL 1
1250	1250	1000		

3.5.1 Tensile strength. The stud shall develop the minimum axial load specified in table III when tested in accordance with 4.7.1.

TABLE III. Minimum Tensile Strength Of Studs (Nut End)

NOMINAL SIZE (NUT END)	TENSILE STRESS AREA mm^2 1/	MINIMUM AXIAL LOAD (kN) 2/	
		LOAD RATING PER TABLE II	
		900 MPa	1250 MPa
M5x0.8	14.0	12.6	17.5
M6x1	19.8	17.8	24.7
M7x1	28.5	25.6	35.6
M8x1.25	36.1	32.5	45.1
M8x1	38.8	34.9	48.5
M10x1.5	57.3	51.6	71.6
M10x1.25	60.6	54.5	75.7
M12x1.5	87.2	78.5	109.0
M12x1.25	91.3	82.2	114.0
M14x1.5	123.5	111.0	154.0

1/ The tensile stress areas used for the calculation of the axial load values are based on the stress area per formula

$$A_S = \pi \left(\frac{E}{2} - \frac{3H}{16} \right)^2$$ as shown in FED-STD-H28, APPENDIX A5, 3. Strength Factors

2/ The minimum axial loads shown are the applicable tensile stress areas multiplied by the appropriate load rating.

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3.5.2 Proof load. The stud shall be capable of withstanding proof loads as specified in table IV when tested in accordance with 4.7.1.

TABLE IV. Proof Load For Studs (Nut End)

NOMINAL SIZE (NUT END)	TENSILE STRESS AREA mm^2 1/	MIN PROOF LOAD (kN) 2/ PROOF STRENGTH PER TABLE II	
		630 MPa	1000 MPa
M5X0.8	14.0	8.8	14.0
M6X1	19.8	12.5	19.8
M7X1	28.5	18.0	28.5
M8X1.25	36.1	22.7	26.1
M8X1	38.8	24.4	38.8
M10X1.5	57.3	36.1	57.3
M10X1.25	60.6	38.2	60.6
M12X1.5	87.2	55.0	87.2
M12X1.25	91.3	57.5	91.3
M14X1.5	123.5	77.8	123.0

1/ The tensile stress areas used for the calculation of the axial load values are based on the strees area per formula

$$A_S = \pi\left(\frac{E}{2} - \frac{3H}{16}\right)^2$$

as shown in FED-STD-H28, Appendix A5, 3. Strength Factors

2/ The minimum proof loads shown are the applicable tensile stress areas multiplied by the appropriate proof strength.

3.5.3 Resistance to pullout. The stud shall demonstrate minimum shear engagement areas as specified in table V. The installed stud with lockring in place as specified in 4.1.3 shall have a minimum resistance to pullout in accordance with table VI. Values specified in table VI are based on the minimum engagement areas of the stud (table V) when installed in a test block of 170 MPa shear strength and tested in accordance with 4.7.2.

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TABLE V. Shear Engagement Area Of Studs

NOMINAL SIZE (NUT END)	MIN SHEAR ENGAGEMENT AREA (mm^2) 1/	
	COL 1	COL 2
	-100 NUMBERS	-200 NUMBERS
M5	73.1	127.8
M6	93.6	172.2
M7	140.4	253.1
M8	203.3	356.5
M10	324.6	559.0
M12	504.6	858.4
M14	643.4	1123.8

1/ Shear engagement area is the assembled dimensional value for the overall engaged area of mating thread members. It does not represent a dimension of either of the members in an unassembled condition.

TABLE VI. Resistance To Pullout, Stud End Threads

NOMINAL SIZE (NUT END)	MIN RESISTANCE TO PULLOUT (kN) 1/	
	COL 1	COL 2
	-100 NUMBERS	-200 NUMBERS
M5	12.4	21.7
M6	15.9	29.3
M7	23.9	43.0
M8	34.6	60.6
M10	55.2	95.0
M12	85.8	146.0
M14	109.0	191.0

1/ Installed in a test block with a shear strength as specified in 3.5.3.

3.5.4 Rotation resistance. The stud shall produce a minimum torque resistance value not less than that specified in table VII when installed in conformance with the applicable specification sheet and tested in accordance with 4.7.3.

3.5.4.1 Removal and replacement. The stud and lockring shall meet the requirements of 3.5.4 when replacing an identical stud and lockring in an existing hole in accordance with the applicable specification sheet provided the hole is undamaged by the stud removal operation.

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TABLE VII. Rotational Resistance Strength

NOMINAL SIZE (NUT END)	MINIMUM ROTATIONAL RESISTANCE (TORQUE-OUT) N-m
M5	7.0
M6	10.0
M7	17.5
M8	28.0
M10	54.0
M12	90.0
M14	155.0

3.5.5 Hardness. The stud shall meet the hardness as specified in the applicable specification sheet when tested in accordance with 4.7.4.

3.6 Metallurgical properties.

3.6.1 Discontinuities. The stud shall not exhibit discontinuities which exceed the following limitations when tested in accordance with 4.8.

3.6.1.1 Cracks. (See 6.3.1) The stud shall be free of cracks in any direction or location.

3.6.1.2 Laps and seams. (See 6.3.2 and 6.3.3) The stud shall show no evidence of laps and seams as specified in 3.6.2. Discontinuity depths shall not exceed the values specified in table VIII.

TABLE VIII. Discontinuity Depth

NOMINAL SIZE (NUT END)	DISCONTINUITY DEPTH MAX mm 1/
M5 THRU M8	0.13
M10	0.15
M12, M14	0.2

1/ Depth of discontinuity shall be measured normal to the surface at a point of greatest penetration.

3.6.1.3 Inclusions. (See 6.4) The stud shall show no evidence of surface or subsurface inclusions at the thread root when examined in accordance with 4.8.

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3.6.2 Thread discontinuities. (Lap, seams, and surface irregularities in rolled threads). The stud shall show no evidence of laps at the root or along the flanks of the thread as shown in figure 2. Multiple laps on the sides of threads are not permissible regardless of location. A single lap is permissible along the side of the thread above the pitch diameter on either the pressure or non-pressure side (one lap per thread) provided it extends toward the crest and generally parallel to the side as shown in figure 3a. Crest craters, crest laps, or a crest lap in combination with a crest crater are permissible, provided the imperfection does not extend deeper than 20 percent of the basic thread height (See Table IX) as measured from the thread crest when the thread major diameter is at a minimum size (See Figure 3a). Slight deviation from the thread contour is permissible at the crest of the thread within the major diameter limits as shown in Figure 3b. The incomplete threat at each end of the thread may also deviate slightly from contour.

TABLE IX. Basic Thread Height, Metric Thread

PITCH mm	BASIC THREAD HEIGHT (.750H)	20% BASIC THREAD HEIGHT
0.8	0.600	0.120
1.0	0.750	0.150
1.25	0.938	0.188
1.5	1.125	0.225

3.6.3 Grinding burns. The stud shall show no evidence of grinding burns.

3.7 Workmanship. Workmanship shall be consistent with the type of product, finish, and class of thread fit specified. The studs shall be of uniform quality free from laps, cracks, seams, inclusions, splits, or other defects or irregularities which would be detrimental to the performance of the stud in service use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.1.2 Test block fabrication. Test blocks shall be fabricated in accordance with figures 4 or 5 as applicable. Test blocks may exceed dimensions specified in figures 4 or 5 to accommodate multiple testing of studs and lockrings except for tests of 4.7.1 and 4.7.2.

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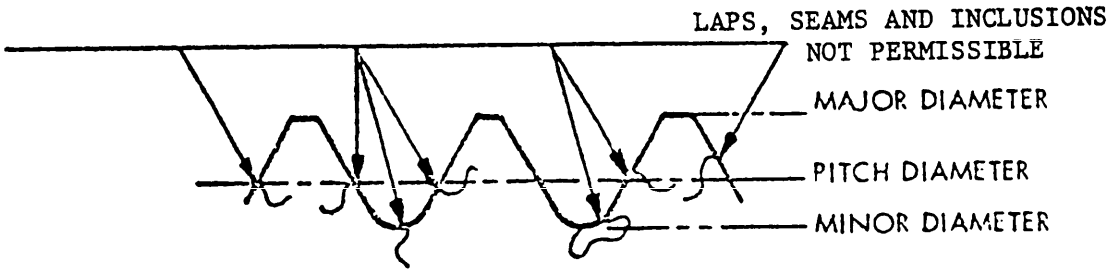


FIGURE 2. LAPS, SEAMS, AND SURFACE THREAD

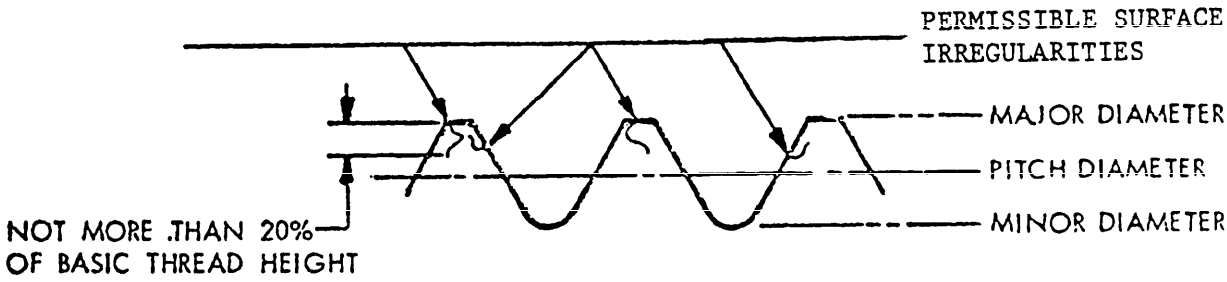


FIGURE 3a. LAPS, SEAMS, AND SURFACE THREAD

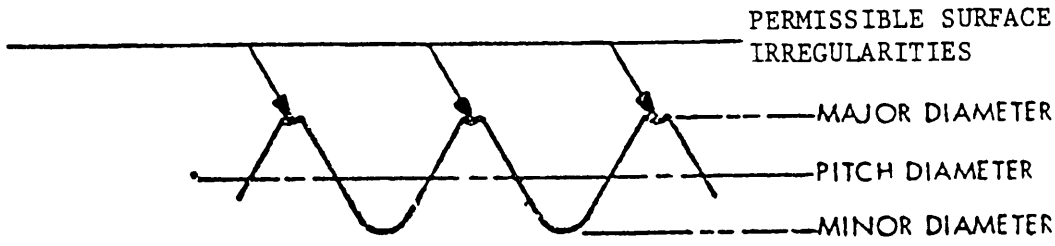


FIGURE 3b. LAPS, SEAMS, AND SURFACE THREAD

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4.1.3 Test specimen installation. Sample studs and lockrings shall be installed in accordance with the applicable specification sheet in test blocks specified in 4.1.2.

4.1.3.1 Inspection of installed studs and lockrings. The studs and lockrings installed as specified in 4.1.3 shall be visually inspected under 10 diameters magnification. The presence of cracks in either test block or stud as a result of installation shall be cause for rejection. When visual inspection discloses a condition which shows cause for further examination, the test specimen shall be penetrant inspected in accordance with MIL-I-6866.

4.2 Quality conformance inspection. Quality conformance inspection shall be as specified in table X.

4.2.1 Inspection lot. An inspection lot shall consist of all studs covered by a single specification sheet produced under essentially the same conditions and offered for inspections at one time.

4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. A resubmitted lot shall be inspected using tightened inspection. Such lots shall be separate from new lots and clearly identified as reinspected lots.

4.3 Classification of defects. Classification of defects shall be in accordance with MIL-STD-105 and table XI.

4.3.1 Defect noncompliance. A stud exhibiting one or more defects shall be considered defective.

4.4 Inspection sampling. Inspection sampling shall be in accordance with MIL-STD-105 and the applicable inspection level and Acceptable Quality Level (AQL) specified in table XI.

4.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with requirements of PPP-H-1581.

4.6 Methods of inspection.

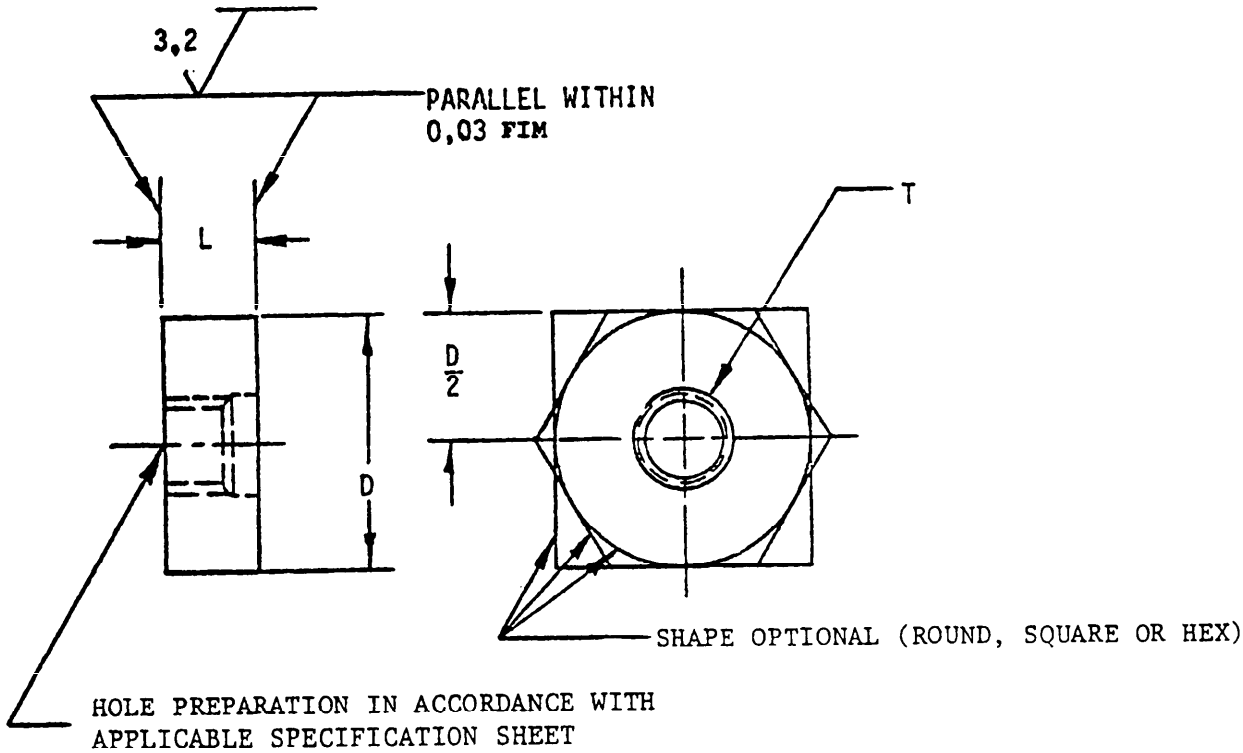
4.6.1 Visual and dimensional. The stud shall be examined to verify that physical dimensions, grinding burns, locking serrations, surface texture and workmanship are in accordance with the applicable requirements of 3.4, 3.4.2, 3.4.3 and 3.6.3.

4.6.2 Material inspection. Material inspection shall consist of certification supporting verifying data that the materials used in fabricating the stud are in accordance with the applicable requirements of 3.2.

4.6.3 Treatment and finish inspection.

4.6.3.1 Heat treatment inspection. Heat treatment verification of the stud shall be in accordance with MIL-H-6875 and 3.3.1.

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1. DIMENSIONS:

   T = Nominal Thread Diameter of Applicable Stud (STUD END).
   D = 4 x T (For T Thread Sizes Smaller Than M12).
       3 x T (For T Thread Sizes M12 and Larger).
   L = Length of Stud End Plus 1,6 For Rotational Resistance Tests (3.5.4).

2. MATERIAL:

   Aluminum Alloy, 2024-T4 Per QQ-A-225/6 (BAR).

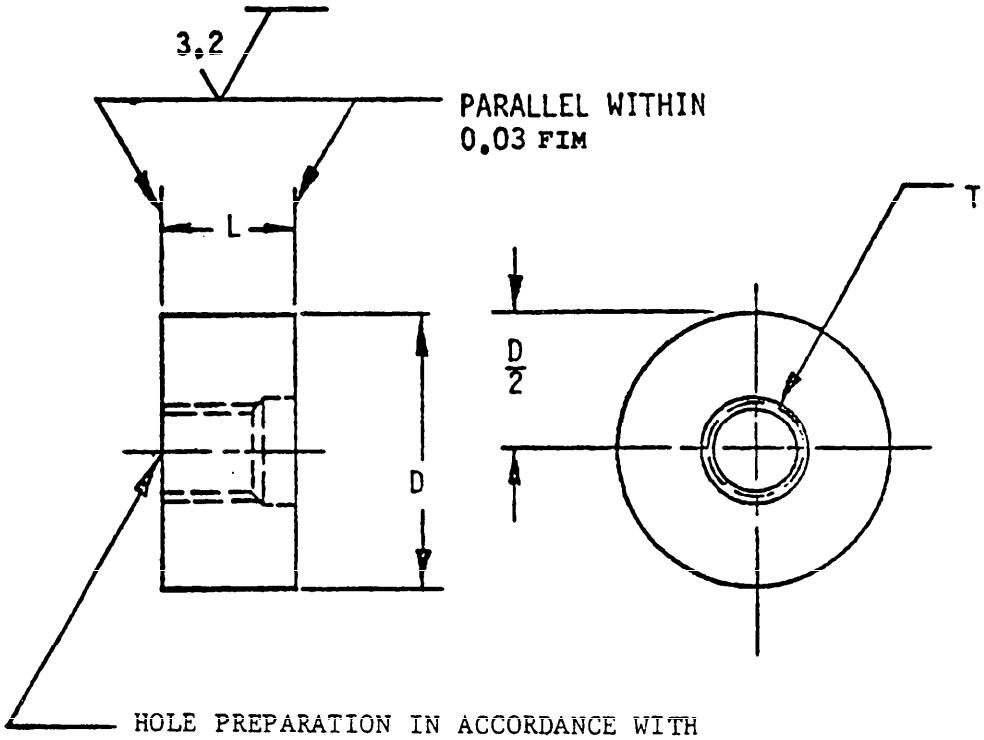
3. Anodize Per MIL-A-8625, Type I, Class 1 or Class 2.

4. Penetrant Inspect Per MIL-I-6866: Discontinuities or Cracks Not Acceptable.

FIGURE 4.

ROTATIONAL RESISTANCE
(TORQUE-OUT)
TEST BLOCK

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1. DIMENSIONS:

   T = Nominal Thread Diameter of Applicable Stud (STUD END).
   D = 4 x T (For T Thread Sizes Smaller Than M12).
       3 x T (For T Thread Sizes M12 and Larger).
   L = Length of Applicable Stud End Plus 1.6.

2. MATERIAL:

   Aluminum Alloy 5083-H321 Per QQ-A-250/6 (PLATE)

3. Anodize Per MIL-A-8625, Type I, Class 1 or Class 2.

4. Penetrant Inspect Per MIL-I-6866: Discontinuities or Cracks Not Acceptable.

FIGURE 5.

PULLOUT TEST BLOCK

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TABLE X QUALITY CONFORMANCE INSPECTION

Inspection	Requirement Paragraph	Test Method Paragraph
Group A		
Visual & Dimensional	3.1	4.6.1
Material	3.2	4.6.2
Treatment and finish	3.3(as applicable)	4.6.3 (as applicable)
Grinding burns	3.6.3	4.6.1
Workmanship	3.7	4.6.1
Packaging	5.1	4.5
Group B		
Threads	3.4.1	4.6.4
Locking serrations	3.4.2	4.6.1
Surface texture	3.4.3	4.6.1
Straightness	3.4.4	4.6.5
Group C		
Mechanical properties	3.5	4.7
Tensile strength	3.5.1	4.7.1
Proof load	3.5.2	4.7.1
Resistance to pullout	3.5.3	4.7.2
Rotation resistance	3.5.4	4.7.3
Removal and replacement	3.5.4.1	4.7.3
Hardness	3.5.5	4.7.4
Discontinuities	3.6.1	4.8
Cracks	3.6.1.1	4.8
Laps and seams	3.6.1.2	4.8
Inclusions	3.6.1.3	4.8
Thread discontinuities	3.6.2	4.8

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Table XI Classification of defects & inspection sampling

Category	Defect	Inspection Method
Critical	None defined	
MAJOR	AQL = 1.5 percent defective, Level II	
101	Threads not as specified (3.4.1,3.4.1.1, 3.4.1.2)	Standard Inspection Equipment (SIE)
102	Shank diameter (3.4)	SIE
103	Incomplete threads (3.4.1.4)	SIE
104	Grip length (3.4)	SIE
105	Stud end length (3.4)	SIE
106	Drilled hole in nut end missing (when required) (3.4)	VISUAL
107	Concentricity between shank and locking serrations (3.4.2)	SIE
108	Imperfect serrations (3.4.2)	VISUAL
109	Straightness of stud (3.4.4)	SIE
110	Surface texture (3.4.3)	SIE
111	Treatment and finish (3.3 as applicable)	VISUAL
112	Thread discontinuities (3.6.2)	SIE
MINOR A	AQL = 2.5 percent defective, Level S1	
201	Overall length (3.4)	SIE
202	Drilled hole diameter and location (3.4)	SIE
203	Workmanship (3.7)	VISUAL
MINOR B	AQL = 4.0 percent defective, Level S1	
301	Chamfer on thread ends (3.4)	VISUAL

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4.6.3.2 Protective finish inspection. Samples studs shall be inspected for minimum thickness and continuity of plating in accordance with QQ-P-416 and 3.3.2.1.

4.6.3.3 Cleaning and descaling inspection. Sample studs shall be inspected for cleaning and descaling in accordance with ASTM A380 and 3.3.2.2.

4.6.4 Thread inspection.

4.6.4.1 Nut end threads. Sample studs nut end threads shall be inspected in accordance with system 23, FED-STD-H28/20 and 3.4.1.1.

4.6.4.2 Stud end threads. Sample stud end threads shall be inspected in accordance with system 23 FED-STD-H28/20 and 3.4.1.2.

4.6.5 Straightness inspection. Sample studs shall be inspected for straightness when rolled on a surface plate. The clearance measured by a feeler gauge shall not exceed the values specified in 3.4.4.

4.6.6 Surface texture inspection. Sample studs shall be inspected for surface texture with any of the surface examination and measurement methods specified in ANSI B46.1 in accordance with the requirements of 3.4.3.

4.7 Mechanical properties inspection.

4.7.1 Axial strength (tensile and proof) inspection. Nut end threads of the studs shall be tested in tension between the nut end and the stud end in accordance with ASTM E 8. Samples shall be of sufficient length to develop the full strength of the nut end thread of the stud without stripping the thread. The axial tensile load values determined by table III and specified in table IV as applicable, shall be applied by the test assembly. The test may be terminated when proof load is reached providing there is no evidence of failure of the nut end or stud end threads. Studs not having sufficient length for tensile tests shall be accepted on the basis of hardness test results.

4.7.2 Resistance to pullout test. Sample studs shall be installed in test blocks in accordance with figure 5 and 4.1.3 to meet the requirements of 3.5.3. The minimum single shear strength of the test block material, shall be determined by means of double shear tests of coupons taken from the material from which the test blocks are fabricated. The bushing and test block clearance holes in the upper and lower yokes (See Figure 6) shall have a free fit not in excess of 1.5 greater than the bushing and test block diameters. The stud clearance hole in the upper bushing (See Figure 6) shall be 0.13 to 0.4 diameter larger than the nominal shank diameter of test stud specimen. The stud clearance hole in lower yoke shall have a diameter of 1.5 x the nominal stud end thread diameter $\pm$ 0.4. The studs shall be tested with a threaded member of sufficient size to develop the full strength of the nut end of the stud without stripping the thread (See Figure 6). The bearing face of the threaded member shall be located a minimum of two and a maximum of three pitches from the stud thread termination. Studs having a grip length of less than twice their shank diameter need not be axial tested. To demonstrate average shear engagement area, an axial load, of magnitude sufficient to produce failure (pullout) shall be applied to the assembly.

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The axial load test result shall be adjusted for test block shear strength as follows: The product of the axial load test result (kN) and the ratio of 170 MPa to the actual minimum shear strength shall be used as the resistance to pullout (N). The resultant value should compare favorably with Table VI. Rate of loading shall not exceed 700 MPa per minute per square mm of the minimum shank area of the stud. Whenever the pullout values exceed the tensile strength of the stud or the test loads exceed the capabilities of the test equipment, it is permissible to use lower shear strength materials (100 MPa approx.) in lieu of test block materials specified in 4.1.2. Double shear tests on coupons taken from the lower shear strength test blocks shall be performed and the resultant shear strength used to adjust the test result as specified above.

4.7.3 Rotational resistance test. Sample studs shall be installed in test blocks in accordance with figure 4 and 4.1.3 to meet the requirements of 3.5.4 and 3.5.4.1. Torque-out values, with no axial load on the stud, shall not be less than the values specified in table VII. The test can be accomplished with a stud removal tool which shall be positioned on the stud shank or threaded portion of the nut end. The rotational force shall be applied by a torque wrench in a counter-clockwise direction.

4.7.4 Hardness test. Sample studs shall be tested for hardness to meet the requirements of 3.5.5 in accordance with MIL-STD-1312, test 6.

4.8 Discontinuities inspection. Sample studs shall be inspected for discontinuities by magnetic particle inspection performed in accordance with MIL-I-6868 for alloy steel. Penetrant particle inspection performed in accordance with MIL-I-6866 for corrosion resistant steel shall be used to determine the presence of cracks and dissontinuities such as laps, seams and inclusions. Magnetic or penetrant particle indications alone shall not be cause for rejection. If indications are considered cause for rejection, representative samples shall be taken from those studs showing indications and these samples shall be further examined. Studs may be sectioned as shown in Figure 7 and shall be inspected parallel to the axis. Discontinuities are measured microscopically under 10X magnification to determine conformance to the requirements of 3.6.1 and 3.6.2. The inspection shall be performed on finished studs free of lubrication and subsequent to any processing operation which could adversely affect the studs. Requirement for dye as an indication of particle inspection may be waived. Particle inspection shall not be required for studs having a nut end diameter of less than 6mm.

4.9 Grain flow inspection. Sample studs specimens shall be micro-examined to determine compliance with the requirements of 3.4.1.5 grain flow for rolled threads. Specimens shall be taken from the finished stud as shown in Figure 7. The studs shall be etched in a suitable etchant for sufficient time to reveal the macrostructure property.

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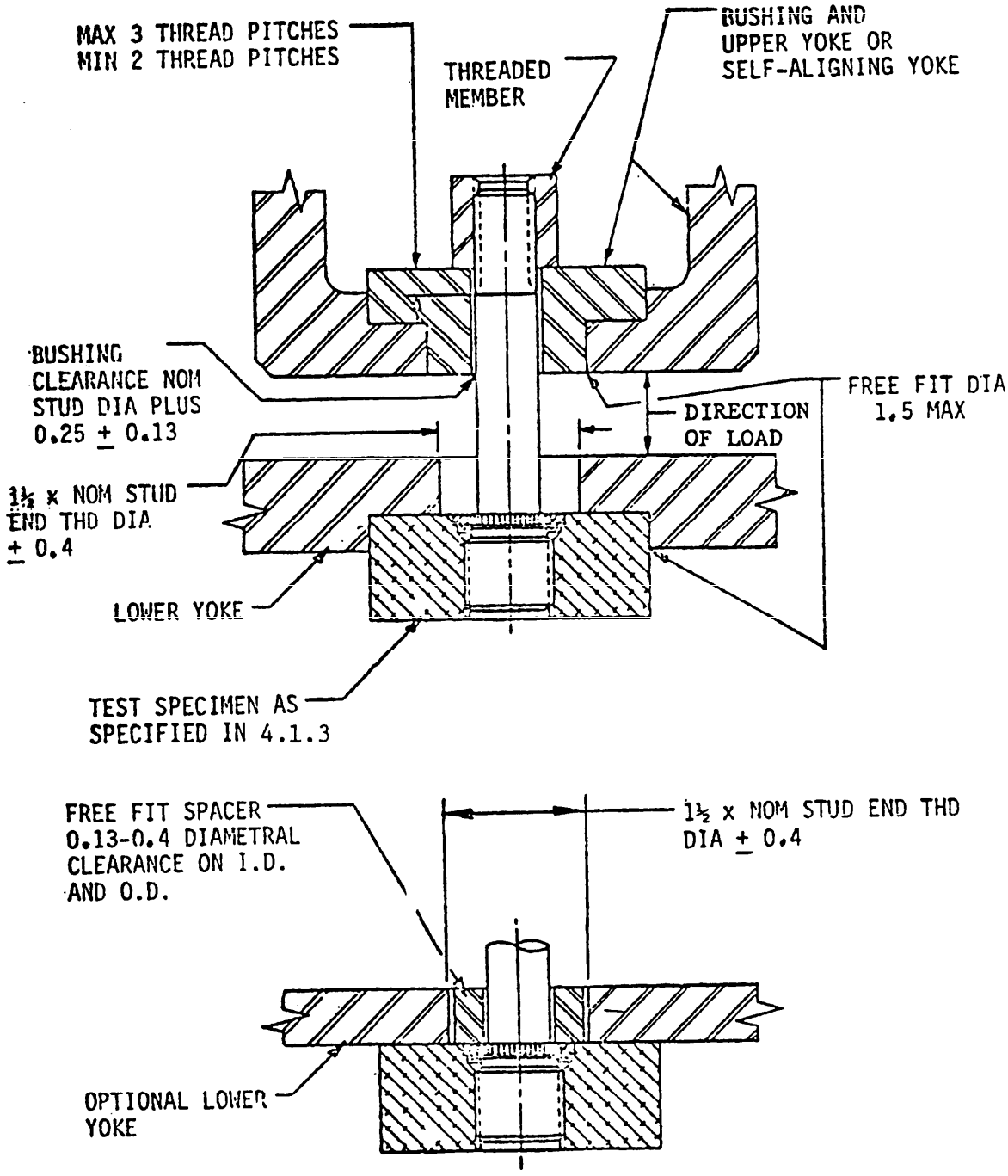


FIGURE 6.
RESISTANCE TO PULLOUT FIXTURE (3.5.3)

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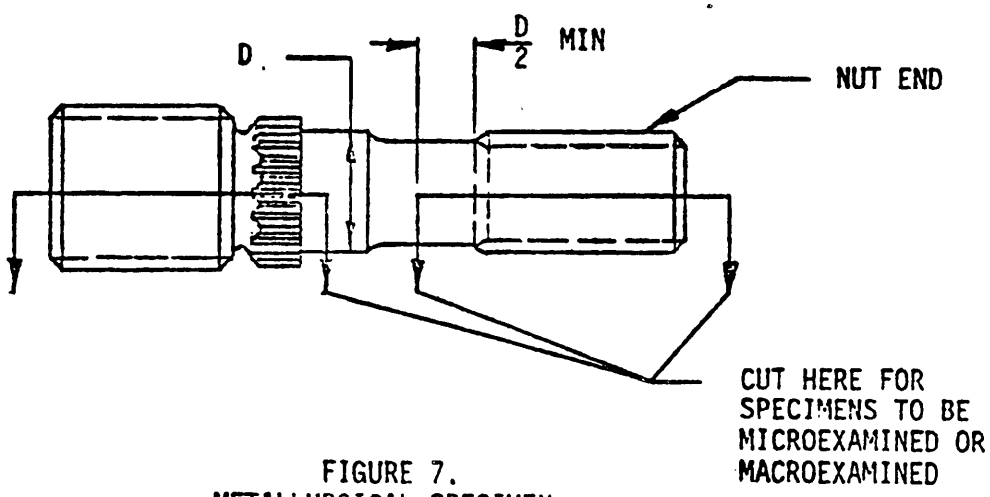


FIGURE 7.
METALLURGICAL SPECIMEN

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (See 6.2).

6. NOTES

6.1 Intended use. Studs covered by this specification are intended as a general-purpose metric fastener with a mechanical lock feature to resist rotation.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification and the applicable specification sheet.

b. Applicable specification sheet part number (See 3.1).

c. Level (degree) of protection, in accordance with PPP-H-1581, ordering data (See 5.1).

6.3 Definitions.

6.3.1 Crack. A crack is a clean crystalline break passing through the grain or grain boundary without the inclusion of foreign elements.

6.3.2 Lap. A lap is a surface defect appearing as a seam, caused by the folding over of metal fins or sharp corners and then rolling or forging them into the surface, but not welding them.

6.3.3 Seam. A seam is an unwelded fold or lap which appears as an opening in the raw material as received from the source.

6.3.4 Inclusions. Inclusions are non-metallic materials in a solid metallic matrix.

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6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

DOD-S-63275A Stud, Screw Thread, Locked In, Ring Locked, Serrated, Metric and Ring, Lock, Serrated, Metric

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